

HEALTHCARE OPERATIONS & FINANCIAL PERFORMANCE

Business Conclusion Report

End-to-end analysis: SQL Server → Python (EDA + Statistics) → Power BI

Prepared by: Mir Sufiyan Ali

Dataset: healthcare_dataset.csv | 54,966 admission records analyzed
August 8, 2026

Executive Summary

This report analyzes 54,966 hospital admission records (55,500 raw rows, 534 duplicates removed) covering patient demographics, billing, admissions, and clinical outcomes across a simulated multi-hospital network. The pipeline moved data from Python cleaning into SQL Server for business-question querying, then into Power BI for an operational dashboard, with formal statistical testing layered on top.

The headline finding is not a growth story — it's a data-quality one. Across nearly every categorical field (hospital, doctor, insurer, admission type, test result), the distribution is close to statistically flat. That pattern, combined with 39,876 distinct hospital names across 54,966 rows, is a strong signal that this is a synthetic, randomly generated dataset rather than real operational history. The numbers below are accurate calculations on the data as given; treat hospital/doctor "leaderboards" as illustrative of the pipeline, not as real business signal.

54,966 Admission Records Analyzed	\$1.40bn Total Billing (Revenue)	\$25,544 Avg. Billing per Patient
15.5 days Avg. Length of Stay	534 Duplicate Records Removed	106 Negative Billing Entries (flagged)

Objectives

- Establish baseline operational and financial metrics (age, billing, length of stay).
- Identify which conditions, hospitals, doctors, and admission types drive cost and stay duration.
- Statistically test whether admission type and gender relate to cost, stay, and admission pattern.
- Package findings into a Power BI dashboard for ongoing monitoring.

Dataset Description

Source: healthcare_dataset.csv — 55,500 rows × 15 columns (patient name, age, gender, blood type, medical condition, admission/discharge dates, doctor, hospital, insurance provider, billing amount, room number, admission type, medication, test results). No nulls were present in any column.

Cleaning steps (Python)

- Standardized name casing (title case) and column names (lowercase, snake_case).
- Removed 534 exact-duplicate rows (55,500 → 54,966).
- Parsed admission/discharge dates and derived length_of_stay.
- Loaded the cleaned table into SQL Server as healthcare_data for querying.

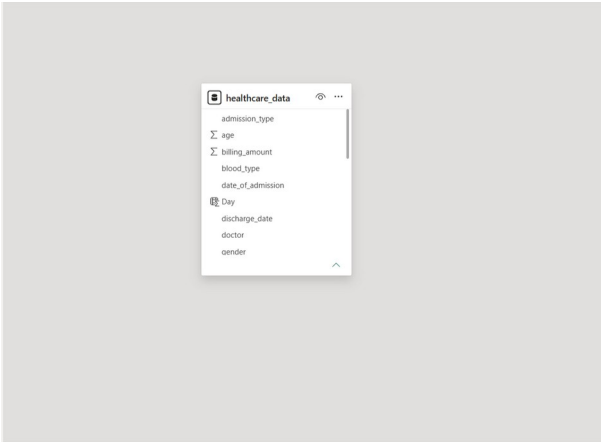


Figure 1 — Data model (healthcare_data table, loaded into Power BI)

SQL Analysis — Selected Business Questions

Business question	Answer
Age group needing most care	65+ (16,096 patients)
Most common medical condition	Arthritis (9,218 patients)
Highest-revenue condition	Diabetes (\$236.5M)
Most frequent blood type	A- (6,898 patients)
Admission type split	Elective 33.6% / Urgent 33.5% / Emergency 32.9%
Peak admission month	August (4,785 admissions)
Most-prescribed medication	Lipitor (11,038 prescriptions)
Lab result split	Abnormal 33.5% / Normal 33.3% / Inconclusive 33.1%

Statistical Analysis

Descriptive statistics

Metric	Age	Billing Amount	Length of Stay
Mean	51.5	\$25,544	15.5 days
Median	52.0	\$25,543	15.0 days
Std. Dev.	19.6	\$14,208	8.7 days
Min / Max	13 / 89	-\$2,008 / \$52,764	1 / 30 days

Distribution & outliers: age, billing, and length-of-stay are all close to uniform (flat) rather than normal or right-skewed — there is no small cluster of extreme high-cost cases driving the average. The one true outlier signal is structural: 106 records carry negative billing amounts (as low as -\$2,008), which is a data-entry/refund-coding issue, not a clinical outlier, and should be corrected at the source before this figure is used in real financial reporting.

Correlation: age, billing amount, and length of stay are all essentially uncorrelated with one another (r between -0.00 and 0.01). This directly means longer stays are not associated with higher bills in this dataset — worth flagging because it runs counter to the intuitive assumption (and to an earlier draft note in the underlying notebook) that stay length drives cost.

Hypothesis testing

Test	Result	Conclusion
T-test — Billing: Emergency vs. Elective	$t = -0.72$, $p = 0.472$	Fail to reject H_0 — no significant billing difference
ANOVA — Length of stay across admission types	$F = 2.00$, $p = 0.135$	Fail to reject H_0 — no significant stay difference
Chi-square — Gender $\times$ Admission type	$\chi^2 = 9.81$, $p = 0.007$	Reject H_0 — statistically significant, but the counts differ by <3% (9,281 vs 9,192 elective; similarly close for the others) — not a practically meaningful gap

Power BI Dashboard

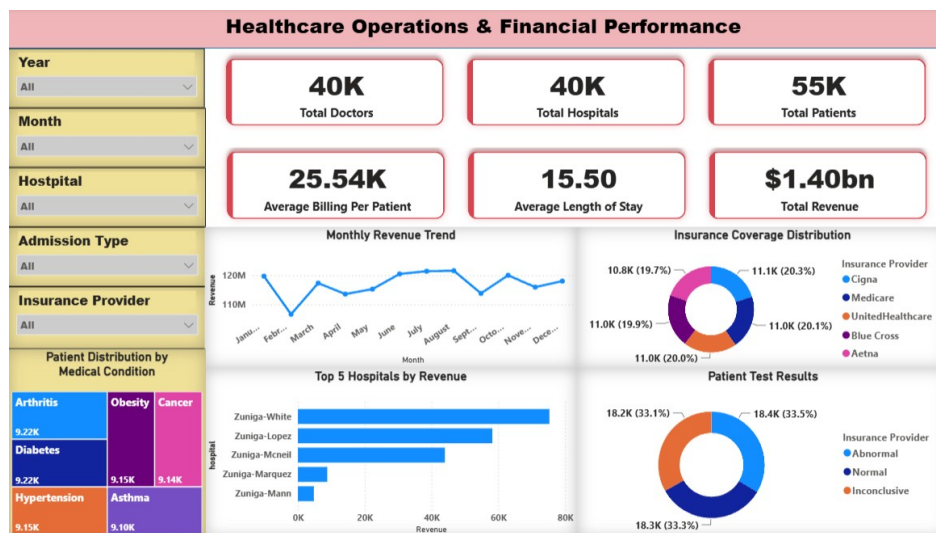


Figure 2 — Healthcare Operations & Financial Performance dashboard

The dashboard mirrors the flat-distribution pattern from the statistical tests: insurance coverage splits ~20% per provider, test results split ~33% each, and the top-5 hospitals by revenue are separated mainly by chance rather than a real performance gap.

Key Business Insights

- No meaningful correlation exists between age, billing, and length of stay ($r \approx 0.00-0.01$) — cost is not being driven by how long or how old the patient is, at least in this data.
- Emergency admissions do not cost significantly more than elective ones, and length of stay does not significantly differ by admission type — two assumptions worth re-testing on real data before acting on them.
- Hospital, doctor, insurer, and test-result distributions are all near-uniform, and 39,876 of 54,966 rows have a unique hospital name — strong evidence this is a synthetic dataset, so hospital/doctor "top performer" rankings should not be treated as real operational findings.
- 106 negative billing entries (down to -\$2,008) indicate a data-quality gap that would need fixing before this pipeline is trusted for real financial reporting.

Data quality note

- 534 duplicate rows removed pre-analysis; source had no null values.
- Gender $\times$ admission-type association is statistically significant ($p=0.007$) but the effect size is too small to be practically actionable.

Business Recommendations

- Use this project to demonstrate the SQL $\rightarrow$ Python $\rightarrow$ Power BI pipeline and statistical methodology — not as a source of real operational claims, given the uniform/synthetic nature of the data.
- Before deploying against live hospital data, add a validation rule rejecting negative billing amounts (or routing them to a refund/adjustment table instead).
- Re-run the correlation and hypothesis tests on real admissions data — the "no relationship" findings here are a property of this dataset, not a general healthcare-costs conclusion.
- Add hospital/doctor IDs (not just names) so entities can be tracked reliably across records.

Future Improvements

- Source de-identified real-world claims data to replace the synthetic set.
- Automate data-quality checks (duplicates, negative billing, null rates) directly in the SQL layer.
- Track patients longitudinally across multiple admissions rather than one row per visit.
- Extend hypothesis testing to length-of-stay by condition and readmission risk once longitudinal data is available.

Conclusion

The pipeline itself — cleaning, SQL analysis, statistical testing, and dashboarding — is sound and reusable. The findings from this particular dataset are limited by the data's synthetic, uniformly distributed nature: none of the tested relationships (age, billing, stay, admission type) show a meaningful real-world pattern, and the one statistically significant result (gender $\times$ admission type) is too small to act on. The clearest next step is applying this same pipeline to real hospital data, where the same tests are far more likely to surface genuine cost and utilization drivers.